

Review Article

The role of hybrid renewable energy systems in covering power shortages in public electricity grid: An economic, environmental and technical optimization analysis

Yasser Fathi Nassar^{a,*}, Hala Jarallah El-Khozondar^{b,c}, Masoud Ali Fakher^d

^a Wadi Alshatti University, Brack, Libya

^b Islamic University of Gaza, Gaza, Palestine

^c Department of Materials and London Centre for Nanotechnology, Imperial College London, United Kingdom of Great Britain and Northern Ireland

^d Engineering Faculty, Wadi Alshatti University, Libya

ARTICLE INFO

Keywords

Hybrid renewable energy system
PV
Wind energy
Pumped hydropower storage
Optimal sizing
Libya

ABSTRACT

Libya has had a chronic electricity shortage since 2011, which grew to 32.5 % in 2023 and is expected to continue into 2024. The Libyan General Electricity Company was consequently compelled to impose up to eight-hour daily power outages. Challenges in the energy sector's three stages—fuel supply, generation efficiency, and electricity transmission and distribution—are the root causes of this deficit. Decentralizing power generation and switching to renewable energy sources are the main ways to solve this issue. Libya's 25-year strategy (2025–2050) reflects this approach, with the goal of having over 50 % of the country's energy mix come from renewable sources by the end of the plan. Based on existing energy potential maps, this study suggests a hybrid renewable energy system (HRES) that combines wind, solar photovoltaic (PV), and pumped hydropower storage (PHS). Because it will assist the national grid in addressing the ongoing electricity shortage, the proposed system is innovative in its design. A multi-criteria decision-making process is used to optimize the sizes of the system's components, taking into account variables like CO₂ emissions (ECO₂), levelized cost of electricity (LCOE), payback period (PBP), net present value (NPV), and loss of power supply probability (LPSP). Depending on fluid flow rates, the sensitivity analysis reveals that variations in the turbine and pump efficiencies may introduce errors of roughly 8 % in the energy output of the turbine and 5 % in the energy consumption of the pump. Variations in inverter efficiency, however, can be disregarded in the computations because their impact is <0.2 %. The optimal system size for the particular case study in Libya is 762,161 MWh of PHS, 385 MW of wind, and 6000 MW of PV. This HRES produces 12,029,670 MWh of energy annually, which completely makes up the 9,744,033 MWh electricity shortfall. The load directly uses about 41.74 % of the energy produced, with the remaining 39.26 % being stored for later use in the upper reservoir. With a 90 % efficiency rate for the pump/turbine equipment, the remaining 19 % of energy is wasted during the pumping and generation processes. Achieving the minimum values for LCOE (\$132.1/MWh), NPV (\$19.923 billion), LPSP (0), PBP (7.54 years), and CO₂ emissions savings (11,624 kton) is the foundation of the system's design. This will save \$816.246 million in social costs and an additional \$2.189 billion in fuel.

1. Introduction

Fossil fuels are concentrated in a small number of countries, giving those countries influence over energy supply and pricing, in contrast to renewable energy sources, which are dispersed around the world but fluctuate in availability. Geopolitical unrest and conflicts in areas with abundant fossil fuels exacerbate this problem and endanger the stability

of the world economy. Furthermore, the world's transition to greener energy sources has hastened due to increased worries about climate change. In light of this, hybrid renewable energy systems (HRES), especially those that include biomass, hydropower, wind, and solar technologies, have become essential for attaining energy sustainability [1–4]. Recent figures demonstrate the enormous expansion of the worldwide renewable energy sector, with installed capacities in 2023

* Corresponding author.

E-mail addresses: y.nassar@wau.edu.ly (Y.F. Nassar), hkhonzondar@iugaza.edu (H.J. El-Khozondar), m.fakher@wau.edu.ly (M.A. Fakher).

<https://doi.org/10.1016/j.est.2024.115224>

Received 28 September 2024; Received in revised form 6 December 2024; Accepted 26 December 2024

Available online 4 January 2025

2352-152X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

exceeding 405.5 GW for solar PV, 150.3 GW for bioenergy, 123.1 GW for wind, and roughly 24.0 GW for hydropower [5]. Furthermore, around 97 % of the world's energy storage is currently provided by pumped hydro storage (PHS) [6].

Even though Libya has a lot of potential for renewable energy—1750 kWh/kWp of solar PV energy per year [7], 3855 kWh/kWp of wind energy [8], and PHS $44.275 \text{ GWh}/\text{m}^3_{\text{flow}}/\text{s}$ [9]—renewable energy still makes up <1 % of the nation's overall energy output [10]. To cut emissions, lessen reliance on fossil fuels, and solve generation shortages like those in Libya, initiatives are underway worldwide to build massive hybrid energy systems and integrate them with national grids.

Consistent with this pattern, the Libyan government has declared its intention to include renewable energy sources, namely solar and wind, to account for more than half of the nation's energy production by 2050 [11]. Nevertheless, despite their significant potential for strategic use, little research has been done on hybrid renewable energy systems in conjunction with PHS. By creating a specific database on Libya's solar and wind energy resources and PHS potential in highland regions, this study seeks to assist the country's energy transition. The integration of renewable energy systems, whether or not they include energy storage, is growing quickly on a global scale [12]. Numerous studies have examined the economic and environmental consequences of energy storage technologies [13–24].

By discovering >22,000 off-river pumped hydro sites in Australia with an energy storage potential of 67 TWh, Blakers et al. [6] highlight the significance of PHS in assisting renewable energy systems for energy production. The levelized cost of electricity (LCOE) for a PV/wind/PHS system is around US\$20/MWh, according to their economic modeling of an all-renewable power system for Australia. A completely integrated system may lower LCOE by 10 %, to €50/MWh, according to Bogdanov et al.'s [25] vision for an integrated renewable energy system spanning Europe, Eurasia, and the MENA regions. With range of storage capacity including 1495 GWh of PHS, 1317 GWh of batteries, and 630 GWh of CAES, the research suggested significant potential for of renewable energy, i.e. 1350 GW of PV, 1406 GW of wind, 281 GW of biomass capacity, 316 GW of hydropower.

The economic optimization of a PHS system with large-scale PV/W HRES in Brazil was assessed by Vitorino et al. [26], who discovered that the system could drastically lower operating costs, with an LCOE of €18/kWh and the cost reduction could be about \$14 Million of the total operation cost estimated for the power system capacity. A 27 % decrease in the cost of purchasing electricity was demonstrated by Naval et al. [27] when they evaluated the integration of 5750 MWh of PHS with 396 MW of grid-connected renewable energy, including 860 MW of PV and 465 MW of wind. Their findings also showed that energy storage PHS maximizes renewable power and reducing the risk of curtailment.

A hybrid energy system comprising a 1 MW PV solar field, a 3 MW wind turbines farm, and 63.51 MWh of PHS was proposed by Nassar et al. [28] in Libya. This system would generate 9342 MWh of energy annually and cover 100 % of the nation's 6137 MWh annual load. With a projected LCOE of €13.2/kWh and a \$10.5 million investment need, the project would have a large positive impact on the environment by reducing CO₂ emissions by 4385 tons and saving 20,700 tons of diesel fuel or 1.48 million m³ of NG annually.

Three off-grid systems for rural Bangladesh were examined by Das et al. [29], in which the optimal system had LCOE = \$0.27/kWh, consists of PV/Diesel/PHS. In [30] researchers carried out a multi-criteria study for a PV/wind/PHS system in Portugal and found that the hybrid system, which could cover 72 % of the nation's yearly energy consumption, was the most economical option. It had capacities of 4 MW wind, 5 MW PV, and 0.54 MW PHS.

According to these research, hybrid renewable energy system—especially those that include PV, wind, and PHS (HRES) will be essential to power systems in the future because they increase the sustainability and efficiency of supply chains and electricity production.

This literature evaluation informs the following contributions of the current study:

- The proposed system provides the utility grid with energy to cover the shortage of the electricity.
- The study reveals that the difference in the turbine and pump efficiencies cause significant error thus it should be considered while the effect of variations in inverter efficiency is small and can be ignored.
- Emphasizing the technological and financial feasibility of locally accessible renewable resources.
- Determining the optimal size of grid-connected hybrid renewable energy systems with PHS in order to satisfy load demands.
- The societal cost of CO₂ emissions and a lifecycle CO₂ assessment for all system configurations are included in this thorough economic and environmental analysis.
- Providing information that is applicable not only to Libya but also to other emerging nations dealing with comparable energy issues.

The structure of the article is as follows: A summary of Libya's power infrastructure and climate data is given in Section 2. The optimization procedure is described in Section 3, together with the constraints, objective functions, and the restrictions and hypotheses of the investigation. Section 4 presents results and discusses the findings, and Section 5 draws the key conclusions.

2. Key information on Libya

2.1. Geographical and climate information

With latitudes between 18° N and 33° N and longitudes between 8° E and 25° E, Libya is situated in the center northern region of Africa, which is bounded by the Mediterranean Sea (Fig. 1). Its 1900-km coastline extends from Egypt in the east to Tunisia in the west. In addition, the nation borders Algeria to the west, Chad and Niger to the south, and Sudan to the east.



Fig. 1. Libya political map.

[Source: <https://maps.lib.utexas.edu/maps/libya.html>.]

south, and Sudan to the southeast. With a population of about 6.3 million and a growth rate of 1.4 %, Libya occupies an area of 1,761,060 km². Summer temperatures in the southern desert range from 40 °C to 46 °C, while winter temperatures range from 1 °C to 12 °C. In the northern parts, summer temperatures range from 27 °C to 32 °C.

With an average global horizontal solar radiation of over 2500 kWh/m²/year and >3500 h of sunshine yearly, especially in the central and southern regions (Fig. 2a), Libya is classified as being in the “solar belt,” which is known for its high solar radiation [31,32]. Furthermore, wind speeds of up to 8 m/s are observed in certain mountainous and coastal regions, suggesting a high potential for the production of renewable energy (Fig. 2b) [33,34]. Making use of these resources is crucial to reaching Libya’s sustainable development objectives, protecting the environment, and exploiting natural resources as efficiently as possible—all of which are top priority for the Libyan government [35]. With elevations of about 4000 m above sea level, the mountains around the country’s eastern Mediterranean coast present excellent locations for PHS project development.

2.2. Electrical energy situation in Libya

The 13 power plants that make up Libya’s electrical grid are dispersed based on population density. With contributions from combined cycle plants at 27 %, gas stations at 53 %, and steam stations at 19 %, the overall capacity is 8051 MW. The following fossil fuels are used to generate electricity in Libya: crude oil (6.9 %), natural gas (58.7 %), light oil (22.9 %), and heavy oil (11.5 %). With a capacity factor of 51.5 % and an average efficiency of 23 %, the total amount of electricity generated in 2023 was roughly 38,396.3 GWh, while the demand was 48,140.3 GWh, with a load factor of 66.8 % [36]. However, the energy deficit amounted to 9,744,033 MWh. This required a prolonged power outage, sometimes lasting up to eight hours per day.

The main source of pollution in the nation is the electrical industry, which is predicted to have an emission factor of 1037 kg CO₂/MWh [37,38]. The hourly electricity generation, demand, and shortage in the Libyan power sector for 2023 are depicted in Fig. 3.

A whisker plot of Libya’s monthly energy situation profile in 2023 is shown in Fig. 4. Ignoring the blackout that happened on August 9 at 12:00 owing to a lack of fuel, the monthly average energy production is approximately 4129 MWh, with a peak of 6355 MW and a minimum of 1980 MWh. The load varies from a low of 2720 MWh to a maximum of 8234 MWh and an average of 5495.5 MWh every month. The average monthly shortfall is 1348 MWh, the peak is 2108 MWh, and the minimum is 96 MWh.

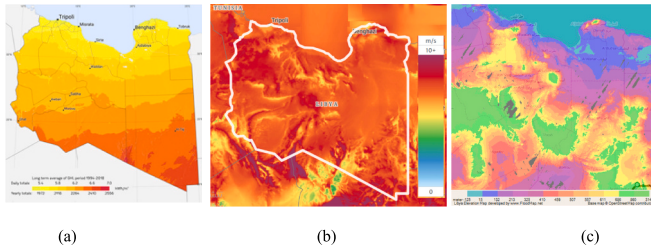


Fig. 2. (a) Annual average global horizontal solar irradiation, (b) wind speed at 100 m, and (c) PHS potential.

[Sources: (a) <https://solargis.com/maps-and-gis-data/download/libya>; (b) <https://globalwindatlas.info/area/Libya/>, (c) https://www.floodmap.net/elevation/CountryElevationMap/?ct=LY#google_vignette.]

3. Methodology

3.1. Layout and operating schemes of the proposed hybrid renewable energy system (HRES)

A hybrid energy system includes both RESs and PHS is analysed in order to increase the research work’s viability. Fig. 5 depicts the proposed system’s layout.

The proposed HRES consists of three primary parts, as shown schematically in Fig. 5: the power supply system, which consists of wind turbine farms and solar PV energy fields; the storage system, which is represented by the PHS; and the consuming side, which is represented by the public grid. To address grid stability during high renewable penetration, the hybrid system leverages HPS as a buffer. When wind and solar production exceeds grid demand, surplus energy is directed to pumping operations, storing potential energy for later use. During low renewable production, the stored hydropower is dispatched to stabilize the grid. Control devices are crucial for maintaining grid voltage and frequency stability, which is seen to be the biggest obstacle to integrating more renewable energy systems into the public grid [39].

The current study offers a mathematical model that simulates and guarantees the automation of the storage system in order to absorb the surplus and cover the shortfall as quickly as possible while maintaining grid stability.

3.2. The design of HRES under investigation

The methodology used to develop and optimize the HRES is shown in Fig. 6. The governing equations for the proposed energy systems are included in the following subsections.

3.2.1. Photovoltaic energy (PV)

The real power (P_{PV}) of the PV panel under real operation and climatic conditions (solar radiation H_t and ambient air temperature T_{∞}) is estimated by [31]:

$$P_{PV} = P_{STC} \times DF_{PV} \times \frac{H_g}{H_{STC}} \times [1 + \beta_{p,T}(T_{cell} - T_{STC})] \quad (1)$$

where P_{PV} represents the actual PV power (W) based on operating conditions; P_{STC} indicates the nominal PV power under Standard Test Condition (STC); $DF_{PV} = 88\%$, refers to the derating factor of PV panel; $\beta_{p,T}$ (%/°C) is the power-temperature coefficient; $T_{STC}(25^\circ\text{C})$ and $H_{STC}(1000\text{W/m}^2)$ are the standard test conditions for temperature and solar radiation, respectively. $T_{cell} = T_{\infty} + 0.07H_g$, is The PV temperature [40] where T_{∞} (°C) and H_g (W/m²) are hourly ambient air temperature and global tilted solar irradiance, respectively.

3.2.2. Wind energy

The energy produced by a specific wind turbine may be estimated as [41]:

$$P_{Wind} = \begin{cases} P_{rat} \left(\frac{V_{Z,t} - V_{cut-in}}{V_{rat} - V_{cut-in}} \right) & V_{cut-in} < V_{Z,t} < V_{cut-off} \\ 0 & V_{Z,t} \leq V_{cut-in} \text{ OR } V_{Z,t} \geq V_{cut-off} \end{cases} \quad (2)$$

where: P_{rat} is the wind turbine rated power, $V_{cut-off}$, V_{cut-in} and V_{rat} are the cut-off, cut-in and rated wind speed, respectively; and $V_{Z,t}$ is the wind speed at the wind turbine hub height (h_z) is estimated by the exponent relation as: $V_{Z,t} = V_{0,t} \left(\frac{h_z}{h_0} \right)^{\alpha}$, where h_0 is the reference height, $V_{0,t}$ is the velocity at h_0 and α is the wind shear coefficient and is taken as 1/7 [42].

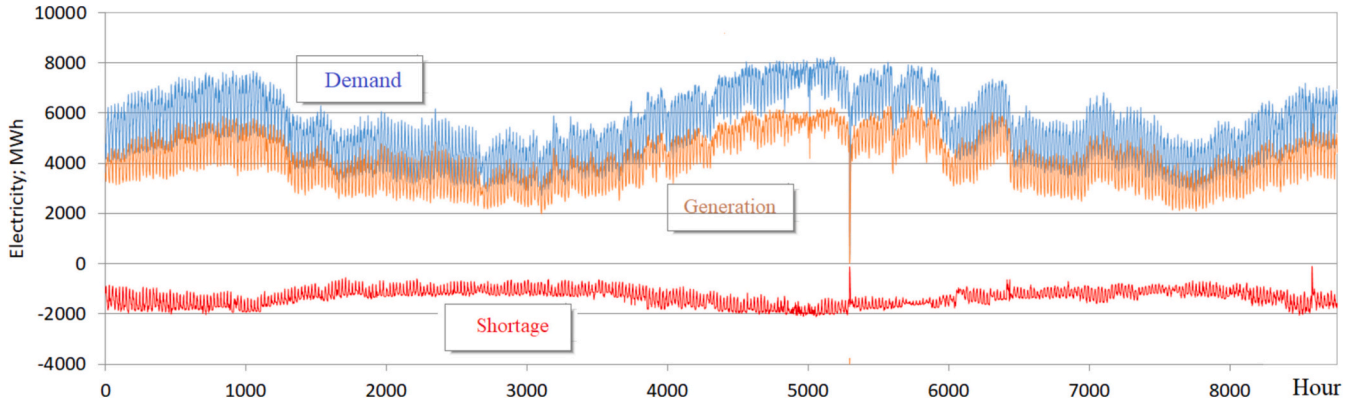


Fig. 3. Hourly energy generated, demand and shortage in Libya.

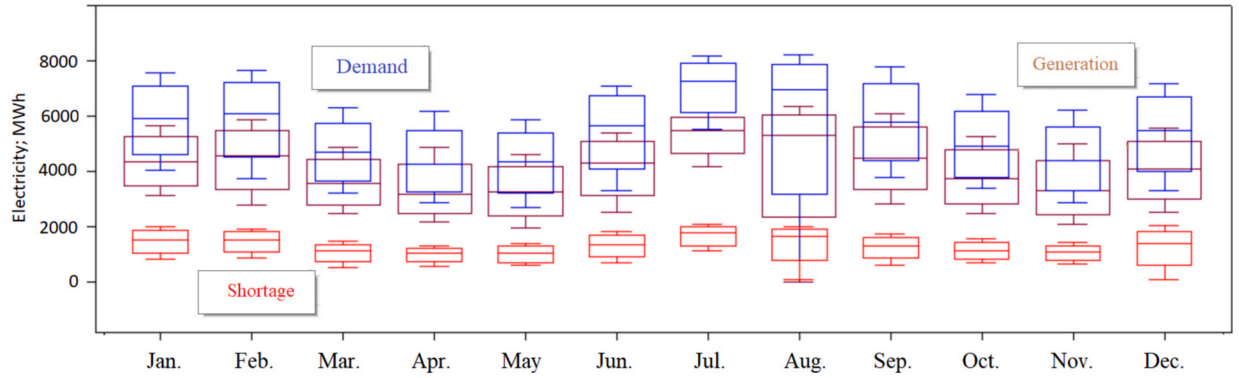


Fig. 4. Statistical analysis of the energy profile in Libya.

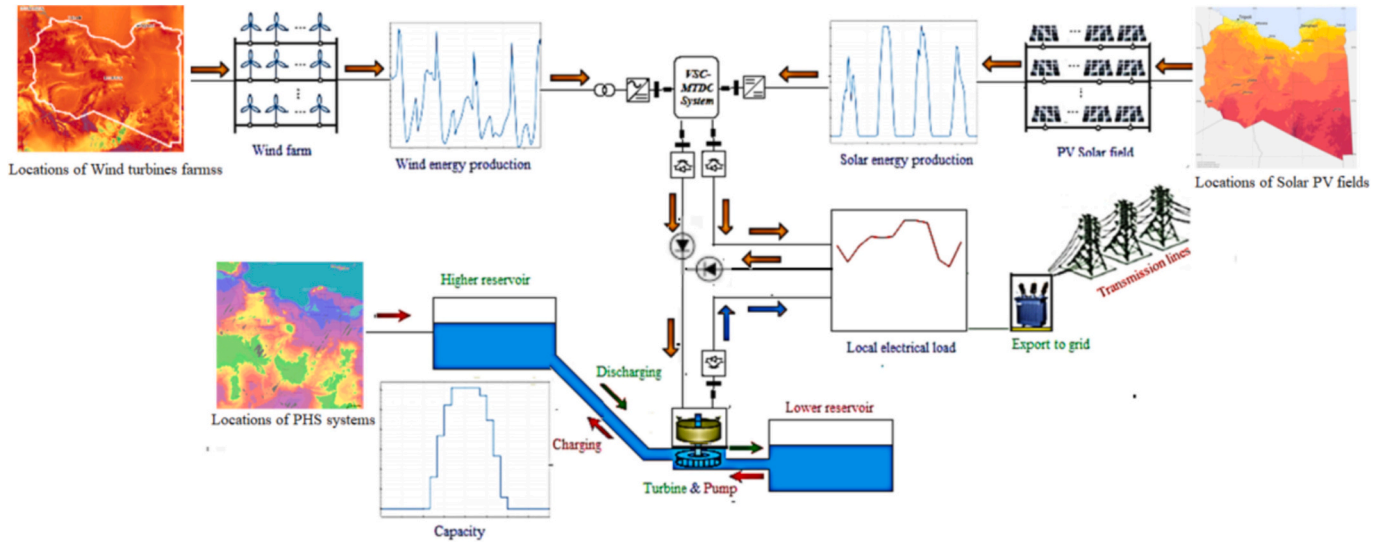


Fig. 5. Layout of the proposed PV/Wind/PHS, promising locations and operating performance curves.

3.2.3. Pumped hydropower storage (PHS)

The power flow in PHS is bidirectional and depends on the operating mode. However, the power—whether in discharging mode (P_d) or charging mode (P_p)—is determined by the parameters of the PHS system as outlined by [43]:

$$P_t = \rho g \dot{V}(H + h_f)\eta_{p/t} \quad (3)$$

$$P_p = \frac{\rho g \dot{V}(H + h_f)}{\eta_{p/t}} \quad (4)$$

where ρ is the water density (1000 kg/m^3), g is the gravity (9.81 m/s^2), $\dot{V}$ is the flow rate (m^3/s), H is the upper reservoir elevation (m), h_f is the friction losses (m) and $\eta_{p/t}$ is the pump/turbine efficiency. The hydrodynamic analysis of the water and penstock design is attached down in

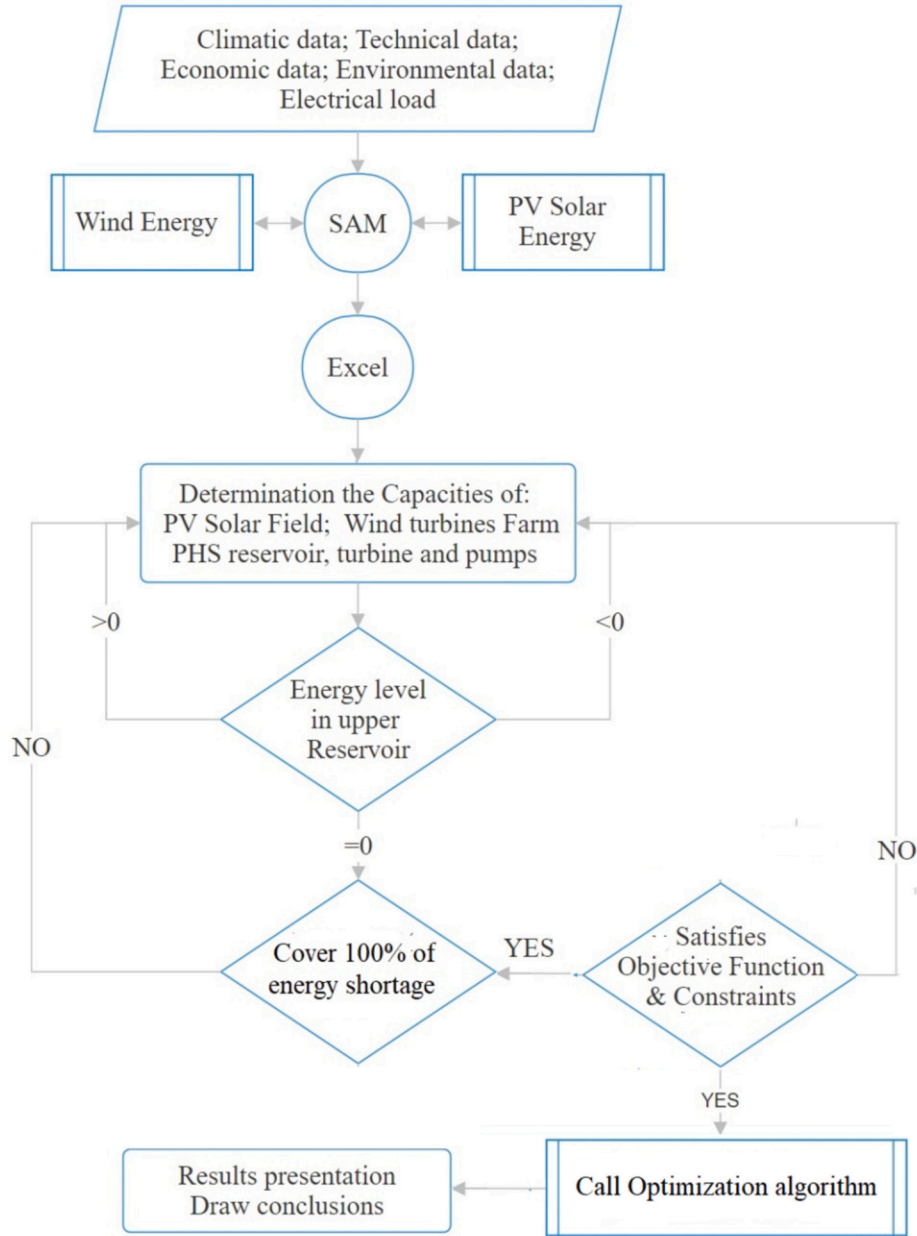


Fig. 6. The followed approach flowchart.

Appendix A.

3.3. The dynamic size of the proposed PHS

Monitoring the energy flow from and to the PHS system, which is connected to the supply system—represented by the wind farm and solar PV field—and the discharge system—represented by the public grid—is required to ascertain the size of the upper reservoir, which is represented by the energy level in the PHS system.

3.3.1. Operation mode of the PHS

The PHS system operates within 3 modes: charging, discharging and standby mode. The following statements describe the atomization regime of the PHS system:

$$\begin{aligned} \text{Charging mode } E_{s-ch}(t) : & [\eta_{inv}E_{PV}(t) + E_w(t)] \\ & > E_{sh}(t) \text{ and } \eta_{p/t} \times [\eta_{inv}E_{PV}(t) + E_w(t) - E_{sh}(t)] + E_s(t) \\ & < E_{ur} \end{aligned} \quad (5)$$

$$\begin{aligned} \text{Discharging mode } E_{s-dis}(t) : & [\eta_{inv}E_{PV}(t) + E_w(t)] < E_{sh}(t) \text{ and } \eta_{p/t} \times E_s(t) \\ & > [\eta_{inv}E_{PV}(t) + E_w(t) - E_{sh}(t)] \end{aligned} \quad (6)$$

$$\text{Standby mode } E_{s-st}(t) : [\eta_{inv}E_{PV}(t) + E_w(t)] = E_{sh}(t) \text{ or } E_s(t) = E_{ur} \quad (7)$$

where: $E_{PV}(t)$, $E_w(t)$, $E_{sh}(t)$ and $E_s(t)$ are the energy generated by the solar PV fields, wind turbines farms, the shortage energy and the energy level in the PHS in MWh at a certain time (t) , respectively, E_{ur} is the upper reservoir capacity in MWh, $\eta_{p/t}$ states for pump and turbine efficiencies, (0.9 [27]) and η_{inv} is the inverter efficiency (0.95 [44]).

3.3.2. Instantaneous PHS energy, $E_s(t)$

The PHS instantaneous energy ($E_s(t)$) is given by,

$$E_s(t) = \eta_{pv/t} \times [\eta_{inv} E_{pv}(t) + E_w(t)] - E_{sh}(t) \text{ and } E_s(t) \leq E_{ur} \quad (8)$$

3.3.3. Storage capacity, E_{ur}

The storage capacity, E_{ur} , stands for the greatest stored energy in the upper reservoir and can be found as follows:

Step 1: The storage capacity's initial value $E_s(0)$ is assumed $= E_{ur}$ where 0 denotes the reservoir capacity's initial iteration.

Step 2: The accumulative sorted energy in the upper reservoir at time t ($E_s(t)$) is calculated for a year's 8760 h.

Step 3: $E_s(t)$ minimum value is calculated. There is a reservoir capacity shortfall if the minimum value is < 0 , and vice versa. When the minimum value of $E_s(t)$ equals zero, that is the ideal value.

Step 4: the minimum value is added to $E_s(0)$ to obtain the actual capacity of the upper reservoir that is sufficient to meet load demand.

Accordingly, the reservoir capacity is found as follows:

$$\begin{aligned} E_s(t + \Delta t) &= E_s(t) + E_s(0) \\ E_s(1) &= \begin{cases} E_s(0) & \text{if } [E_s(t)]_{t=1,8760} = 0 \\ E_s(0) - E_s(t) & \text{if } [E_s(t)]_{t=1,8760} \neq 0 \end{cases} \\ E_{ur} &= \max[E_s(t)]_{t=1,8760} \end{aligned} \quad (9)$$

where Δt is the time increments that equals an hour.

3.4. Objective functions and constraints for optimal sizing of HRES

Four objective functions (OF) are taken into consideration; financial (NPV and LCOE), reliability (LPSP), and CO₂ emissions ($E_{CO_2,HRES}$). Eqs. (10)–(13) represent the objective functions minimization in the aforementioned study as follows:

$$OF_1 \rightarrow \min(NPV) = \frac{(1+d)^n - 1}{d(1+d)^n} \sum_{i=1}^3 [C_{cap-ann} + C_{O\&M-ann} - C_{Env-ann}]_i \quad (10)$$

$$OF_2 \rightarrow \min(LCOE) = \frac{\sum_{i=1}^3 [C_{cap-ann} + C_{O\&M-ann} - C_{Env-ann}]_i}{\sum_{t=1}^{8760} [\eta_{inv} E_{pv}(t) + E_w(t) + E_{S-dis}(t) - E_{S-ch}(t)]} \quad (11)$$

$$\begin{aligned} OF_3 \rightarrow LPSP &= \frac{\sum_{t=1}^{8760} \{E_{sh}(t) - [\eta_{inv} E_{pv}(t) + E_w(t) + E_{S-dis}(t) - E_{S-ch}(t)]\}}{\sum_{t=1}^{8760} E_{sh}(t)} \\ &= 0 \end{aligned} \quad (12)$$

$$OF_4 \rightarrow \min(E_{CO_2,HRES}) = E_{CO_2,PV} + E_{CO_2,W} + E_{CO_2,S} \quad (13)$$

where the real discount rate $d = \frac{i-f}{1+f}$ is calculated using an inflation rate of $f = 2.5\%$ and a nominal discount rate of $i = 6.5\%$ [45], n is lifetime of the system components (25 years for bot PV solar field and Wind turbines farm, and 60 years for PHS equipment). $C_{cap-ann}$ and $C_{O\&M-ann}$ are annual based capital and operation and maintenance costs of the component i , where 1 state for PV, 2 for wind and 3 for PHS equipment. $E_{sh}(t)$, $E_{pv}(t)$, $E_w(t)$, $E_{S-ch}(t)$ and $E_{S-dis}(t)$ are the shortage, PV energy, wind energy, and the energy charged and energy discharged from PHS at a certain time t , respectively. And $E_{CO_2,PV}$, $E_{CO_2,W}$, $E_{CO_2,S}$ are the CO₂ life cycle emission from PV, Wind, PHS respectively.

The annual base capital cost is determined from [46]:

$$C_{cap-ann} = \frac{d(1+d)^n}{(1+d)^n - 1} C_{cap} \quad (14)$$

where C_{cap} is the capital cost for each component. $C_{Env-ann}$ is the

environmental damage coat of CO₂ found as follows [47].

$$C_{Env-ann} = CO_{2savings} \times \phi_{CO_2} \quad (15)$$

where: ϕ_{CO_2} (\$/ton CO₂) is the carbon social cost taken as \$70/ton CO₂ [48].

In order to solve this problem, the optimization process takes into account a number of factors, including the size of the upper reservoirs, the maximum capacity of the inverter, the maximum installation capacity of the PHS pump/turbine, and the proposed HRES's ability to provide a steady supply of energy on its own. This is because it is the only source capable of meeting the nation's load requirements, including solar PV and wind energy capacity. These limitations apply to the previously stated objective functions. The restrictions described in the following inequalities further clarify the optimization problem:

$$\sum_1^{8760} [\eta_{inv} E_{pv}(t) + E_w(t)] > \sum_1^{8760} E_{sh}(t) \quad (16)$$

$$\frac{\eta_{inv} \times P_{inv}}{SF} > P_{rat,PV} \quad (17)$$

$$0 \leq E_s(t) \leq E_S \quad (18)$$

where: $E_{sh}(t)$ is the energy shortage in the grid (MW), P_{inv} is the inverter capacity (MW), P_{rat} is the rated power of HRES components (MW), and SF states for the safety factor (1.2) [49].

3.5. Eco-environmental analysis

The CO₂ life cycle evaluation is the basis of assessing t HRES's environmental effect. As a result, the quantity of CO₂ avoided may be found by subtracting the CO₂ emissions connected with RES equipment's life cycle from the CO₂ emissions of Libya's electricity generation system. The CO₂ amount that is avoided from being emitted into the atmosphere by implementing the planned HRES:

$$CO_{2savings} = EF_{CO_2,Grid} \times E_{sh} - \sum_{i=PV, Wind, PHS} EF_{CO_2,i} \times E_i \quad (19)$$

where: E_{sh} is the annual energy shortage (MWh), E_i represents the annual energy consumed by the load and produced by each component (i), including PV solar energy, wind energy, PHS system in MWh, $EF_{CO_2,Grid}$ (1037 kg CO₂/MWh) is CO₂ emission factor based on life cycle assessment [37], and $EF_{CO_2,i}$ is the CO₂ - life cycle assessment emission factor (kg CO₂/MWh). The CO₂ life cycle assessment emission factors for each component of the HRES are considered as: $EF_{CO_2,PV} = 52$ kg CO₂/MWh [50], $EF_{CO_2,W} = 32$ kg CO₂/MWh [51], and $EF_{CO_2,PHS} = 35$ kg CO₂/MWh [50].

The levelized cost of energy (LCOE) is a complete economic assessment of costs associated with an energy-generating system. It considers: capital expenses, variable and fixed operation and maintenance costs, original investment, and social cost of the CO₂ emissions. The LCOE is computed by dividing the annual value of all costs throughout the system's lifespan by the total electrical energy production. This statistic is especially useful for comparing various energy technologies, which range in lifetime, project scale, capital needs, risks, returns, and capacity. System performance, continuing expenditures, O&M charges, and the amount of air emissions prevented throughout the system's operation all have an impact on LCOE. Reformulating the equations from [52], the LCOE may be expressed as:

$$LCOE = \frac{\left[\frac{d(1+d)^n}{(1+d)^n - 1} (C_{PV} + C_w) + \frac{d(1+d)^{n_{PHS}}}{(1+d)^{n_{PHS}} - 1} C_{PHS} + (O_{PV} + O_w + O_S) - C_{CO_2} \right]}{\sum_{t=1}^{8760} E_{sh}(t)} \quad (20)$$

where C_{PV} and O_{PV} are the capital cost and operational and maintenance cost of the PV system (\$); C_w and O_w are the capital cost and operational

and maintenance cost of the wind turbine system (\$); O_S is the operational and maintenance cost of PHS storage system (\$); C_{CO2} is CO2 saving cost (Eq. (19) including CO2 life cycle assessment of equipment and social cost); n_{PHS} is the life time of the PHS which equals 60 years and n is the renewable energy system life time estimated to be 25 years. In Eq. (21), the total benefits (TB) indicates the HRES's costs over its operating life time of n years of the renewable energy system. To calculate TB for a 25-year project life time with an annual rate of inflation (d), follow the steps indicated in [53]:

$$TB = \left\{ \frac{(1+d)^n - 1}{d(1+d)^n} \right\} I \quad (21)$$

Here, I represents the annual income (\$/year). The average annual benefits TB_{av} , is estimated as:

$$TB_{av} = \frac{TB}{n} \quad (22)$$

Eq. (23) is used to estimate the payback period (PBP):

$$PBP = \frac{C_{PV} + C_W + C_S}{TB_{av}} \quad (23)$$

The cost of PHS system encompasses the expense of a single hydro-power turbine C_t in $(\frac{\$US}{kWp})$, two pumps C_p in $(\frac{\$US}{kWp})$, and the civil structures of penstocks C_{civil} in $\frac{\$US}{kWp}$ and the two reservoirs C_R in $\frac{\$US}{kWp}$. The total cost of PHS equipments can be represented as [28]:

$$C_S = C_t \times P_t + 2C_p \times P_p + \{C_{civil} + 2C_R\} \times P_S \quad (24)$$

where: P_S represents the energy storage capacity of the PHS (kWh) system.

3.6. Optimization algorithm

The optimization procedure has been accomplished through the use of iterative algorithms. The optimization procedure, which uses SAM and Excel software to determine the proper size of RES components, is shown in Fig. 7. The process starts with SAM program input data, then uses Excel to simulate economic and energy performance on an hourly basis. The results are then contrasted with the stated constraints and objectives. For varying component capabilities, this process is repeated. After evaluating the combinations, the design with the lowest LCOE value is deemed ideal.

3.7. Assumptions, limitations of the research and uncertainties of the results

The present study utilizes the following assumptions to streamline the energy-economic-environmental analysis:

- The PHS system is regarded as perfect and lossless without evaporation or leakage.
- The top reservoir's water level is not considered in the head calculation.
- The system's salvage cost is not covered.
- The PHS system's pump/turbine efficiency remains constant regardless of the flow rate.

- Neither the control devices nor the connection terminals experience any energy loss.
- It is assumed that the feed-in tariff of \$0.10/kWh for the grid is equivalent to the price for selling electricity to the local population.

This study's primary drawback is its failure to account for weather-related effects on the higher reservoir's water levels, such as precipitation and evaporation. Furthermore, variations in turbine performance that may arise from different water flow rates in the upper reservoir or variations in pump efficiency brought on by variations in the power supply are not taken into account by the model.

The study's main sources of uncertainty are parameter estimation, model selection, and data availability. The power output profile of the associated systems can be greatly impacted by local climatic conditions, which are a major determinant of the potential of the renewables such as wind and solar energy. Uncertainty is also introduced by the expense of renewable energy initiatives. According to Fathi and Alsadi [54], the cost of water turbines varies from \$353 to \$2216 per kW, whereas the price of PV panels can vary by over 360 %, from \$980 to \$4510 per kW. The operation and maintenance expenses for PHS facilities also vary annually from \$2.12/kWh to \$5.64/kWh.

4. Results and discussion

Finding appropriate sites for the construction of solar energy fields, wind farms, and PHS stations is the first phase because the analysis's main focus is on the potential for solar PV, wind energy, and PHS. The outcomes are divided into the following categories:

1. Estimation of the energy produced by the PHS turbines and the energy provided to the PHS by renewable energy sources (RES).
2. Sizing the PHS system and the supply system (wind and solar PV).
3. Evaluation of the suggested system's environmental effects and economic viability.

4.1. Allocating promising sites for the establishment of solar PV fields, wind turbines farms, and PHS stations

Using the System Advisor Model (SAM), a dynamic simulation tool, the study assessed the technical and financial viability of solar PV, wind, and PHS at twelve chosen locations (Fig. 8). The SolarGis platform's 1-h time series climate data, which spans 13 years from January 1, 2007, to June 30, 2020, was used in the simulations. The results of the simulations are illustrated in Figs. 9–11.

The LCOE and the appropriate PV technology for each location are shown in Fig. 9. The correlation between geographic location, LCOE, and the kind of wind turbine most suited for each site is shown in Fig. 10. The heights of the mountains in the study areas are depicted in Fig. 11, which may be encouraging for the establishment of PHS stations.

In addition to the high potential of the sources (solar energy, wind energy, or elevation), the following criteria have been used to choose sites: proximity to energy transmission lines, proximity to transportation routes, proximity to urban areas, suitability of the land for building the station with low land costs and preparation, high relative humidity, and available water resources (e.g., sea or high capacity wastewater

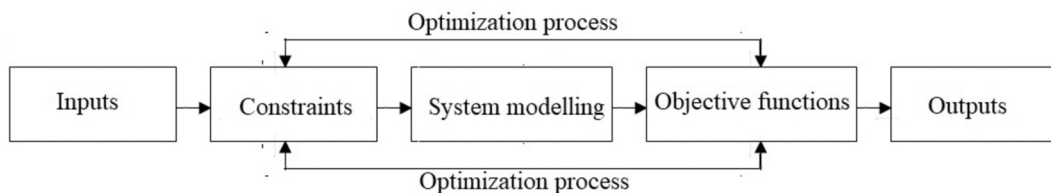


Fig. 7. Optimization process media.



Fig. 8. Map of Libya showing the study areas.
[Source: https://en.m.wikipedia.org/wiki/File:Libya_location_map.svg.]

treatment stations) for PHS stations. As a result, it was determined to divide the project among the regions, as shown in Fig. 12.

As shown in Fig. 12, the methodology in this study will be based on the mean values of wind speed, solar radiation, and mountain heights for each site type assigned to the projects. As a result, Fig. 13 shows the average hourly global horizontal solar irradiance (W/m^2). While Fig. 14 shows the corresponding productivity of 100 MW solar PV field capacity and obtained by SAM.

Fig. 14 embodies the case of PV solar fields in hot regions! At a time when the intensity of solar irradiation is at its highest levels in the summer, it finds that the productivity of PV solar field is at its lowest levels. This is due to increasing the solar cells temperature because of

absorbing more solar irradiation and thus the energy produced from them decreases, as described mathematically by Eq. (1).

While Fig. 15 shows the average hourly wind speed and Fig. 16 displays the corresponding productivity of 100 MW capacity wind turbines farm obtained by SAM.

4.2. Determination of PV solar field and wind turbine farm capacities

As stated in Section 3.4, the power supply system's capacity must be sufficient to provide the PHS with potential energy to meet the energy deficit. Therefore, Eq. (25) can be rewritten in terms of the capacity factors and capacities of the solar PV field and wind turbine farm as follows:

$$CF_{PV} = \frac{\sum_{t=1}^{8760} E_{PV}(t)}{8760P_{PV}} \text{ and } CF_W = \frac{\sum_{t=1}^{8760} E_W(t)}{8760P_W} \quad (25)$$

$$CF_{PV} \times P_{PV} + CF_W \times P_W = \sum_{t=1}^{8760} E_{sh}(t)$$

where: CF_W and CF_{PV} are the capacity factors of wind and PV systems, respectively. Since all the values in Eq. (25) are known except P_{PV} and P_W , a relationship between these two key design parameters can be expressed as $P_{PV} = f(P_W, E_{sh})$ as follows:

$$P_{PV}(P_W, E_{sh}) > \frac{1}{CF_{PV}} \left[\sum_{t=1}^{8760} E_{sh}(t) \right] - \frac{CF_W P_W}{CF_{PV}} \quad (26)$$

The acquired findings are depicted in contour plot in Fig. 17. This graph depicts the relationship between the energy generation side (PV solar field and wind turbine farm) and the energy consumption side (energy shortage). The best allocation of each source in the HRES combination will be established by meeting the design constraints while remaining within the context of goal function stated in Section 3.4.

For the present case study, and from the SAM simulation results, the solar PV capacity factor $CF_{PV} = 20\%$ and the wind energy capacity factor $CF_W = 45\%$. Accordingly, the relation between the capacities (in MW) of solar PV energy and wind energy were found as:

$$P_{PV}(P_W, E_{sh}) = -3.0026P_W + 7094.225 \text{ and } P_W(PV, E_{sh}) = -0.333P_{PV} + 2362.383 \quad (27)$$

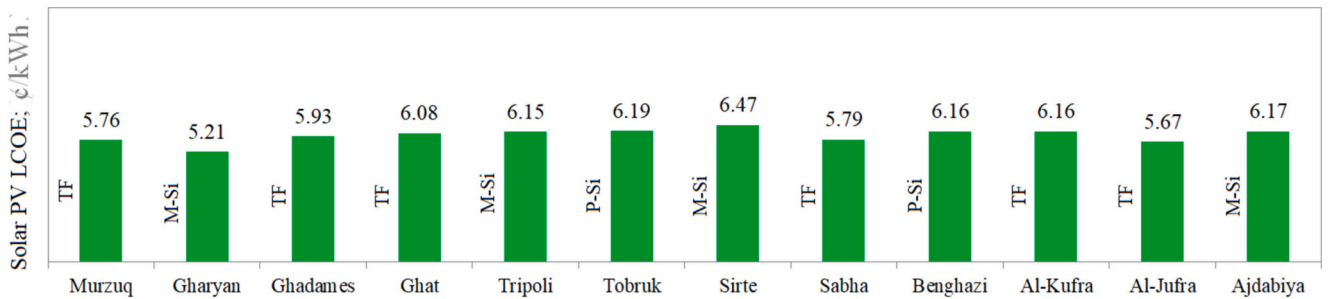


Fig. 9. The LCOE and the perfect technology for Solar PV energy exploitation in several locations in Libya.

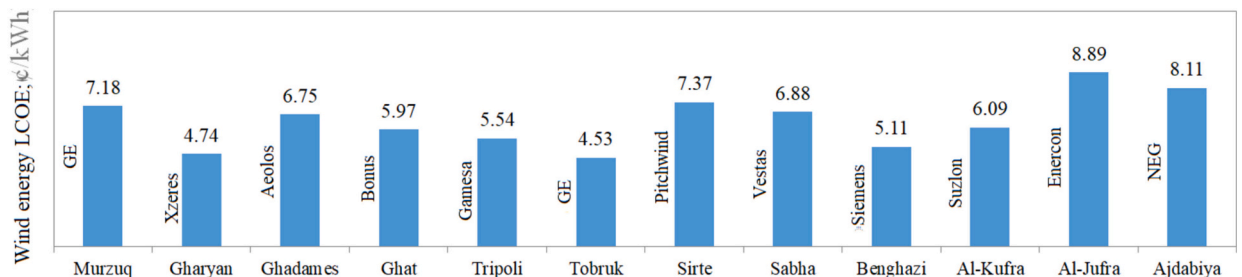


Fig. 10. The LCOE and the perfect technology for wind energy exploitation in several locations in Libya.

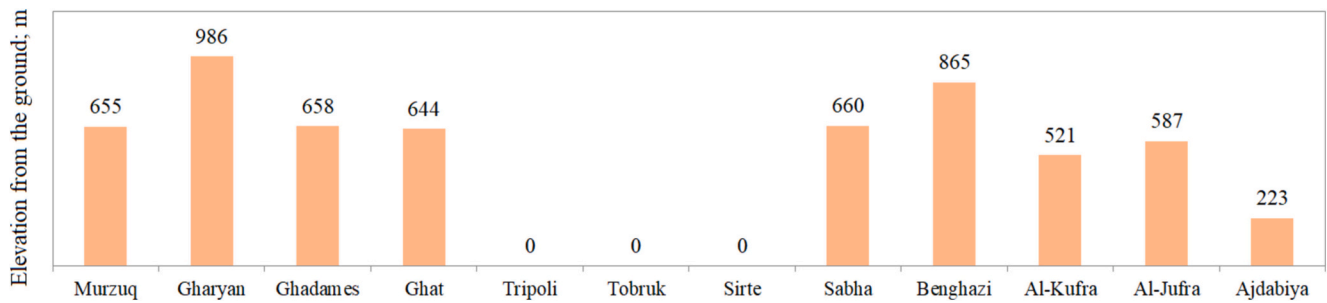


Fig. 11. Elevations of some promising locations for PHS exploitation in Libya.
[Source: <https://www.floodmap.net/?ct=LY>.]



Fig. 12. Map of geographical distribution of the HRES project across all regions in Libya.

4.3. Determination of PHS system capacity

The zero level in the top reservoir is a crucial determinant of the PHS system's size. At this point, the PHS system's necessary size to balance supply and demand will be determined. As described in the methodology section, load data and energy output from different capacities of wind turbine farms and solar energy fields that were created using the SAM program were analysed using Microsoft Excel. Fig. 18, which displays the energy levels in the PHS system corresponding to various capacities of solar PV fields and wind turbine farms, provides a graphic representation of the results.

Fig. 18 illustrates clearly the impact of PHS capacity by the power supply capacities. Fig. 19 shows that wind energy capacity has a bigger impact on the size of the PHS system than solar PV energy. This is mostly due to the load behavior shown in Fig. 3. Solar PV energy output usually coincides with daily peak consumption, which reduces the need for large energy storage. In contrast, wind energy supply frequently peaks at night when demand is low, necessitating storage to satisfy increasing demand later. The needed energy storage capacity was estimated, and the results are depicted as bubble plots in Fig. 19. The charts depict the link between the hybrid system's power capacity and reservoir size (MWh). Additionally, the minimal PHS system size required to overcome power shortages was established.

Fig. 19 clearly shows the inverse link between the generating system (PV/Wind) and the storage system (PHS): as the generation system's capacity grows, the storage system's capacity falls. To investigate the influence of solar irradiation and wind speed on the design parameters of the Hybrid Renewable Energy System (HRES), rising wind speeds, as long as they remain below the cut-out speed of the chosen turbine, increases wind farm output. This minimizes the amount of wind energy required by the hybrid system to fulfill load demand, decreasing the

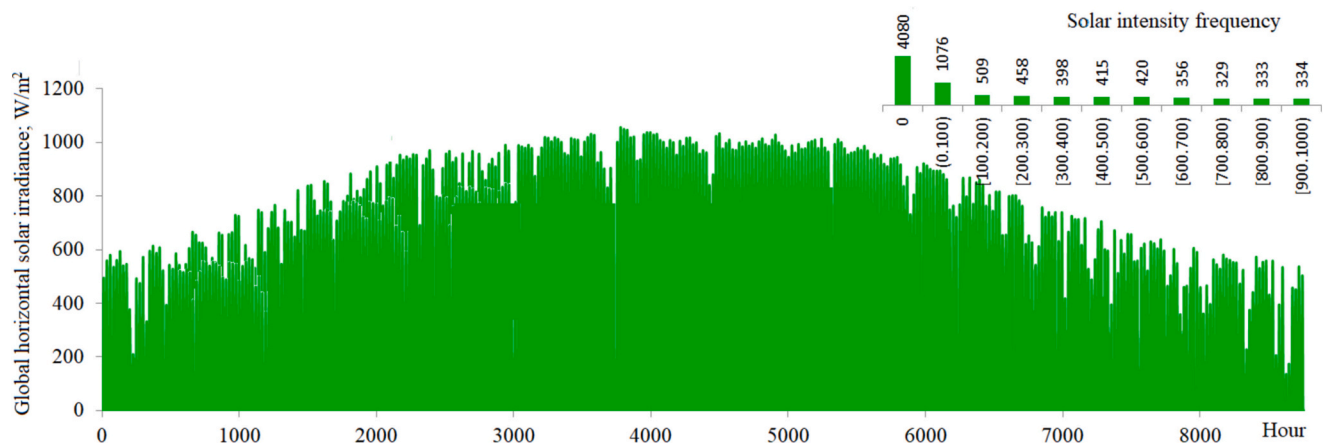


Fig. 13. Average hourly global horizontal solar irradiance (W/m^2) and the frequency of solar intensity.

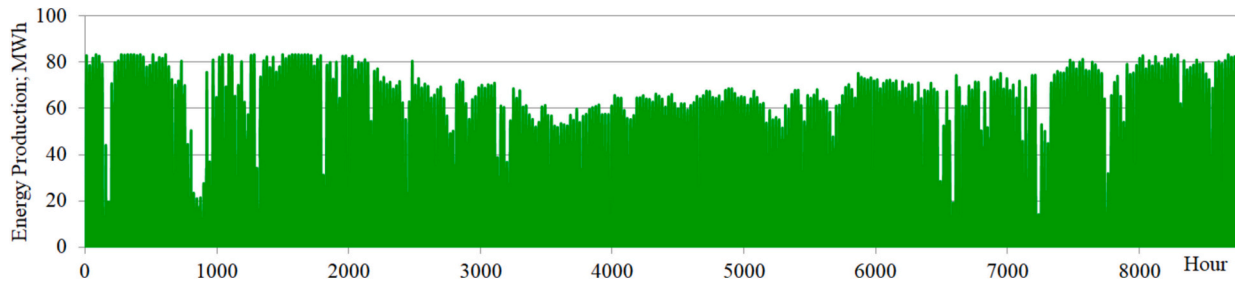


Fig. 14. The energy produced by a 100 MW capacity PV solar field.

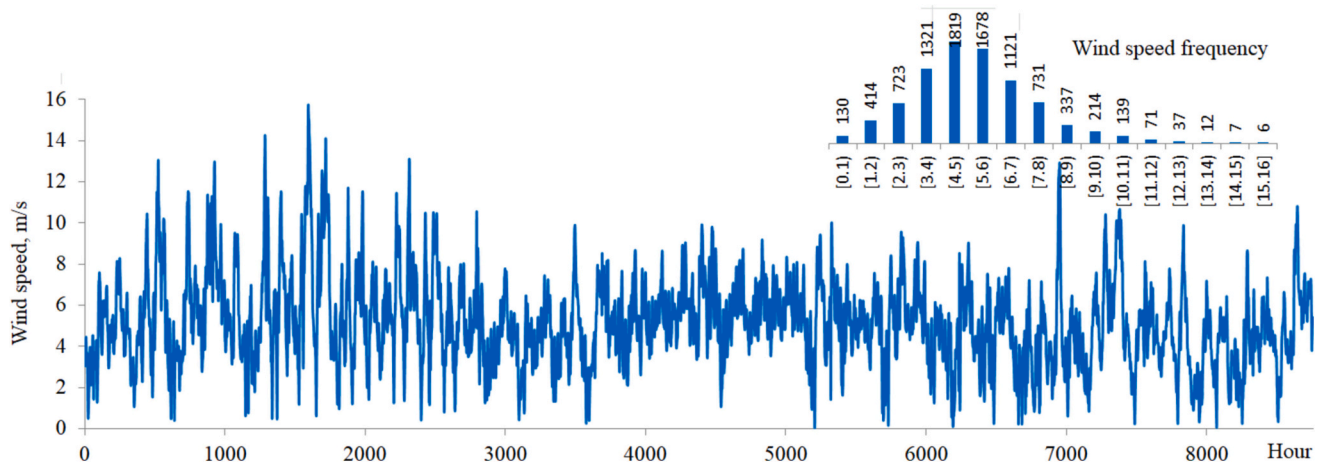


Fig. 15. Hourly wind speed (m/s) and wind speed frequency.

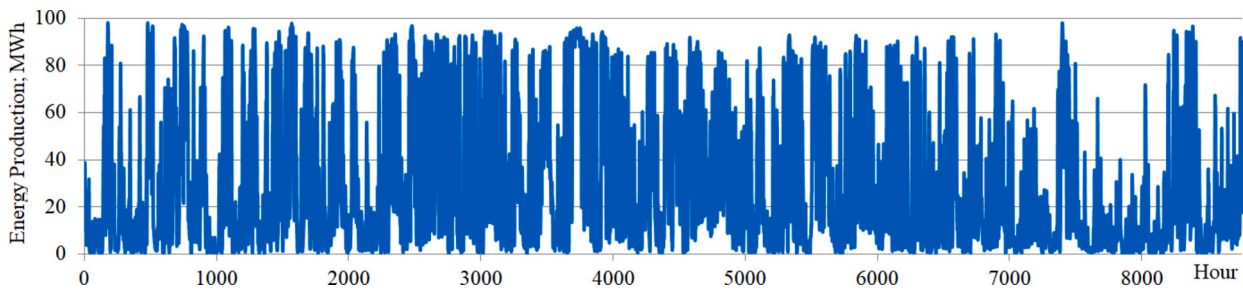


Fig. 16. The energy produced by a 100 MW capacity wind turbines farm.

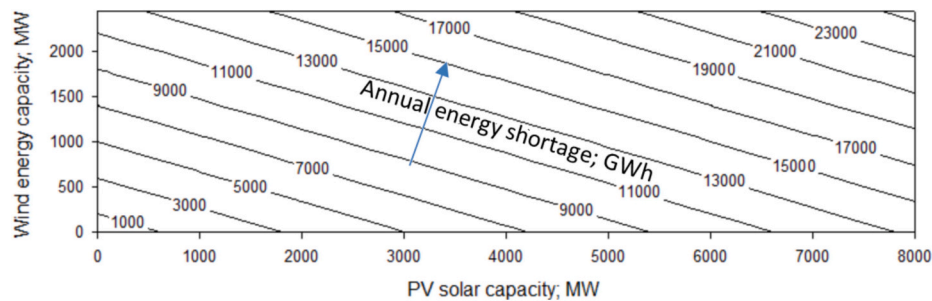


Fig. 17. PV and wind energy capacities requirements.

necessary wind energy capacity. In comparison, the impact of sun irradiation is more complicated. In hot regions, such as Libya, increased sun irradiation raises the temperature of solar cells, lowering their efficiency, as seen in Fig. 14. However, wind acts as a cooling agent, decreasing the temperature of PV modules and increasing their efficiency as wind speeds increase. As a result, increased solar irradiation necessitates an increase in PV solar field capacity to compensate for

lower efficiency.

4.4. Sizing optimization of PV/W/PHS system

The optimal capacity of the suggested HRES was ascertained by applying the objective functions within the specified limitations. As stated in Eq. (26), the optimization process was conducted across a

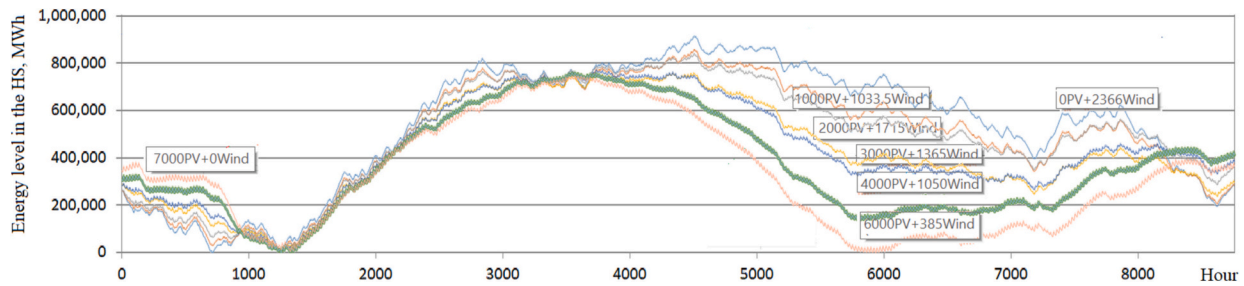


Fig. 18. Energy level in the PHS related to the wind turbines farm and PV solar field capacities.

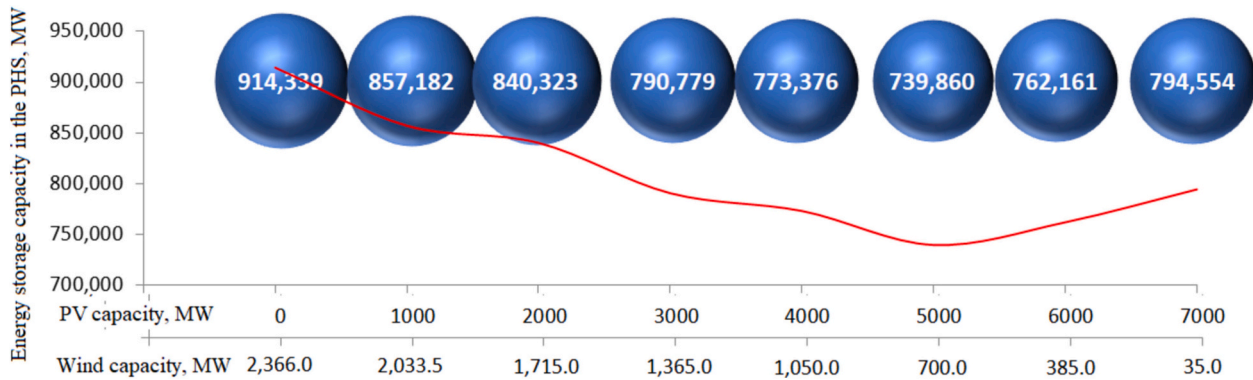


Fig. 19. Sizing of the PHS capacity related to the power supply capacities.

broad range of PV solar and wind energy capacities, with the matching PHS capacity displayed in Fig. 19. The main technical, financial, and environmental information utilized in the system sizing procedure is shown in Table 1 [45,55]. The levelized cost of electricity (LCOE)-based HRES sizing optimization is shown in Fig. 20, which also takes into account the social cost of CO₂ emissions and the CO₂ life cycle analysis of each HRES component.

According to Fig. 20's economic analysis, the minimum LCOE is \$132.12/MWh. Therefore, 762,161 MWh for the PHS, 6000 MW for the solar PV field, and 385 MW for wind energy are the ideal capacity.

Table 1
Economic, technical and environmental data of the HRES components.

Metric	Value
PV solar field [56]	
Capital cost of PV solar field; C_{PV}	\$876/kW
Operation and maintenance; O_{PV}	\$20/kW/year
Life time; n	25 years
Wind turbines farm [57]	
Capital cost of wind turbine farm; C_w	\$1247/kW
Operation and maintenance; O_w	\$81/kW/year
Life time; n	25 year
PHS energy [20]	
Capital cost of PHS; C_{PHS}	\$1982/kWh
Operation and maintenance; O_{PHS}	\$3.88/kW/year
Elevation; H	900 m
Life time; n_{PHS}	60 years
Environmental aspects	
EF_{CO_2} for solar PV energy	52 kg CO ₂ /MWh [50]
EF_{CO_2} for wind energy	32 kg CO ₂ /MWh [51]
EF_{CO_2} for PHS energy	35 kg CO ₂ /MWh [50]
EF_{CO_2} of electricity generation in Libya	1037 kg CO ₂ /MWh [37]
Social cost of CO ₂	\$ 70/ton CO ₂ [48]

4.5. Energy behavior of the proposed HRES

The energy audit of the proposed HRES is shown in Fig. 21, which includes 6000 MW of PV solar field, 385 MW of wind turbines farm and 762,161 MWh of PHS, to meet 9,744,033 MWh annual electrical shortage.

With a 6000 MW PV solar field, 385 MW of wind turbine farm, and a 762,161 MWh storage capacity, the hybrid system generates 12,029,670 MWh of energy annually, which is enough to make up the 9,744,033 MWh electricity shortfall. 39.26 % of the system's total energy is stored as pumped energy in the upper reservoir for later use, while 41.74 % is directly used by the load. The remaining 19.0 % is lost during the generation and pumping operations, with the pumping phase accounting for around 90 % of that loss.

4.6. Economic, environmental and social aspects

The following findings were derived from the CO₂ life cycle evaluation of every component of the proposed HRES as well as the economic and environmental study described in Sections 3.5 and 3.6:

- The project is expected to cost \$28,609,110,002 in capital.
- The net present value (NPV) is \$19,923,445,601.
- The levelized cost of energy (LCOE) is US 0.132/kWh.
- The annual operation and maintenance payments are \$ 8,179,040.
- The annual CO₂ amount savings are 9,497,702 ton CO₂.
- The annual social cost of CO₂ based on (\$70/ton CO₂) is \$ 665,838,140. T
- The annual fossil fuel savings are \$ 2,188,678,304; based on [58].
- The payback money period (PBP) is 11.54 years.
- The annual income from electricity selling (\$100/MWh) is \$974,403,300.

Fig. 22 illustrates the cash flow diagram for the proposed HRES under the Libyan conditions and the prices of September 2024.

The projected HRES's cash flow is shown in Fig. 22. The \$28.610

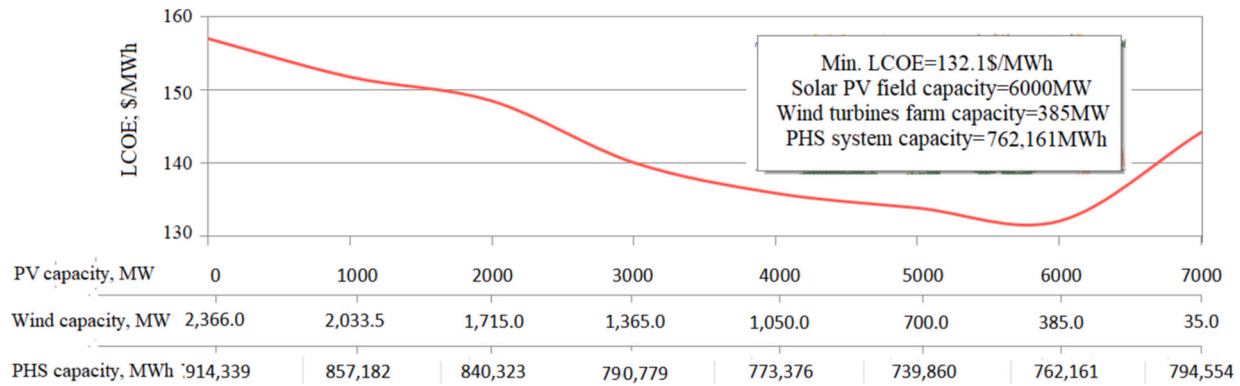


Fig. 20. LCOE optimization of HRES size.

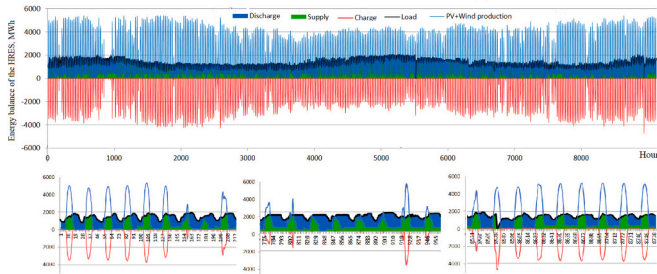


Fig. 21. The energy balance of the proposed HRES.

billion total investment must be made at the start of the project. Since the PHS system has a 60-year lifespan, only the PV/W system's cost, which is \$5.744 billion, needs to be paid after the system's 25-year lifespan. The operation and maintenance expenses require an additional \$187.095 million to be paid each year. Conversely, the HRES generates \$974.403 million, \$665.838 million, and \$2.189 billion in revenue from the sale of energy, the societal CO₂ cost reductions, and the oil cost savings.

4.7. Economic feasibility comparison

When comparing the study's findings to those of other studies throughout the world, numerous important considerations must be addressed. The most important of these are the availability of solar and wind energy resources, as well as LPSP, which is a vital indicator in economic assessments. Table 2 compares technical and economic aspects of numerous past research.

Table 2 shows that the Levelized Cost of Energy (LCOE) varies significantly, which may be ascribed to a variety of design, operational, economic, technological, and climatic elements that influence these systems. A research done in Spain indicated that Pumped Hydro Storage (PHS) is not a viable solution due to present component costs [73].

4.8. Sensitivity analysis

The sensitivity of several design parameters, including equipment efficiencies, upper reservoir volume and elevation [74], power supply loss probability, social costs of CO₂ emissions, and equipment capital costs [20], has been the subject of numerous researches. Research on the effects of dynamic fluctuations in the pump/turbine and inverter efficiency is, nevertheless, scarce. As can be seen in Fig. 23, which shows how the pump/turbine efficiency changes with the percentage of the nominal flow rate, the efficiency of the pump/turbine fluctuates with the fluid flow rate.

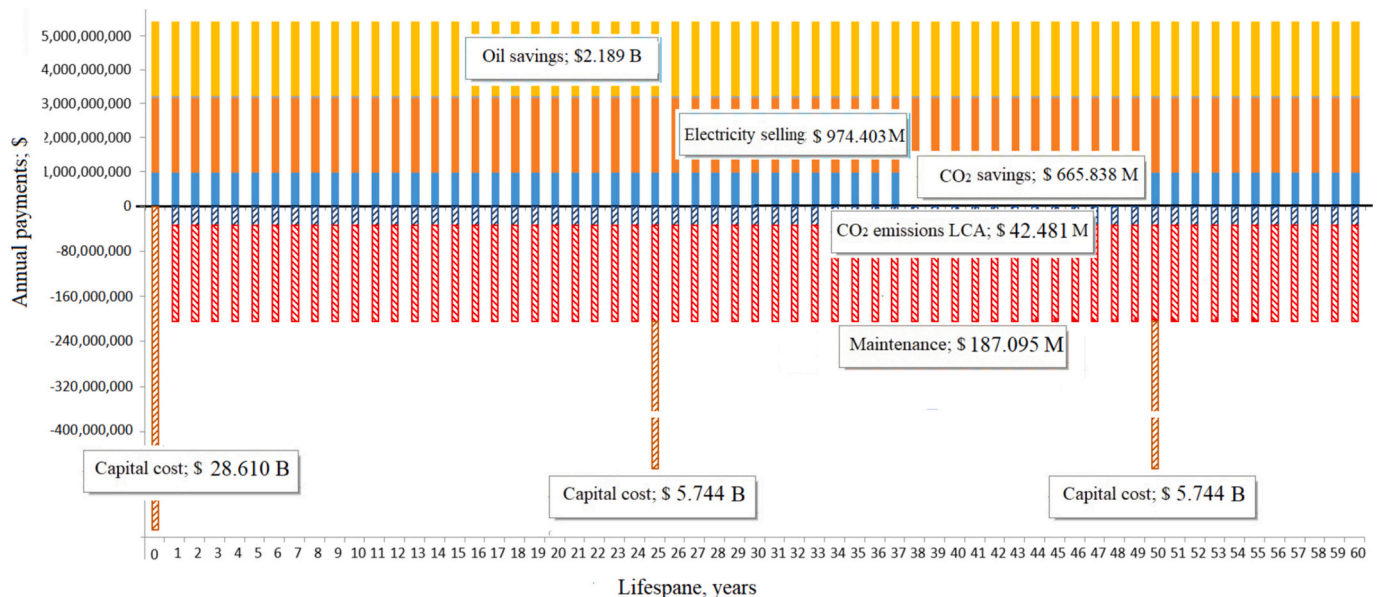


Fig. 22. Cash flow of the proposed HRES.

Table 2

Comparison of PV/Wind/PHS hybrid renewable system LCOE in several locations.

Location	LPSP	PV, MW	Wind, MW	PHS, MW	Grid mode	LCOE (USD/kWh)	Ref.
Brazil	=	0.1	2	0.61	OFF	0.408	[59]
Cameroon	0 %	0.081.8	0.01	0.735	OFF	0.265	[60]
Cape Verde	–	50	40	30	ON	0.23	[61]
China	13.87 %	392.6	120	130	OFF	0.1072	[62]
China	5 %	160	130	90	OFF	0.091	[63]
China	3.3 %	5764	400	400	OFF	0.178	[64]
Egypt	3.296 %	160	130	90	OFF	0.2750	[65]
Hong Kong	0 %	0.119	0	13,205 m ³	OFF	0.289	[66]
India	–	156	108	194	ON	0.4828	[67]
India	–	130	110	60	ON	0.288	[68]
Iraq	2.4 %	43.8	1.5	14,065.72 m ³	OFF	0.1692	[69]
Jordan	–	12.69	0.028	0.107	ON	0.042	[70]
Poland	–	6	12	10	ON	0.142 €	[71]
Portugal	–	1.60	5	144 MWh	ON	0.18 €	[72]
Libya	0 %	6000	385	762,161 MWh	OFF	0.132	This study

The following expression was used to numerically characterize the efficiency performance with great precision ($R^2 = 1$):

$$\eta_{p/t}(NF) = \begin{cases} 0 & \text{if } 0 \leq NF \leq 10\% \\ -26.667NF^2 + 14.0NF - 1.133 & \text{if } 10\% < NF \leq 25\% \\ 0.1443\ln(NF) + 0.9 & \text{if } NF > 25\% \end{cases} \quad (28)$$

where NF is the percentage of the real flow rate divided by nominal flow rate ($\dot{V}$). The turbine only runs at maximum capacity for 760 h (roughly 13.3 % of the total operating hours) while the pump runs at maximum capacity for 227 h (roughly 7.4 % of the total operating hours), according to an analysis of the flow rates during both the charging (pump) and discharging (turbine) modes. On the other hand, the turbine and pump stay in standby mode for roughly 5704 and 3056 h, respectively. The following formula [76] can be used to determine the error ($Er_{np/t}$) in the PHS charge and discharge energy caused by the pump/turbine's missing efficiency:

$$Er_{np/t} = \frac{\sqrt{\sum_{i=1}^n (E_{S,\eta=0.90} - E_{S,\eta=f(NF)})_i^2}}{\sum_{i=1}^n (E_{S,\eta=0.9})_i} \quad (29)$$

Both $E_{S,\eta=0.9}$ and $E_{S,\eta=f(NF)}$ are the estimated power supply (charge mode - pump) or power output (discharging mode - turbine) of the PHS system. They are computed using Eq. (29), which defines a variable efficiency, and a constant pump/turbine efficiency ($\eta = 90\%$). In this instance, there are 8760 data points, or one year's worth of hourly data, denoted by the symbol n . Approximately 8.3 % and 5.9 %, respectively,

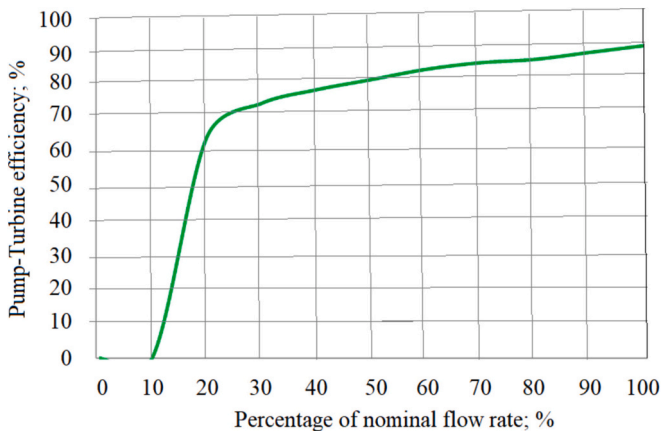


Fig. 23. Pump/turbine efficiency $\eta_{p/t}$ variation with percentage of nominal flow rate (NF) [75].

are the average projected errors caused by the turbine and pump efficiency. The variance in inverter efficiency in relation to the percentage of normalized inverter power is also depicted in Fig. 24.

The power output in relation to the PV solar field's rated power is known as normalized power (NP). The following is a mathematical expression for the inverter efficiency:

$$\eta_{inv}(NP) = \begin{cases} -318.18 NP^2 + 34.363 NP & \text{if } NP \leq 6\% \\ -0.0376 NP + 0.9679 & \text{if } NP > 6\% \end{cases} \quad (30)$$

According to an analysis of the PV solar field's output, it only produced electricity for 383 h at an inverter efficiency of $<95\%$, or $<7\%$ of normalized power, and did not produce electricity for roughly 4652 h. The PV solar field ran at an average efficiency of 95 % during the remaining 3725 h, or $>90\%$ of its total effective hours. It is possible to determine the inaccuracy in the productivity of the PV solar field ($E_{\eta_{inv}}$) caused by changes in inverter efficiency depending on power input using [76]:

$$E_{\eta_{inv}} = \frac{\sqrt{\sum_{i=1}^n (E_{PV,\eta=0.95} - E_{PV,\eta=f(NP)})_i^2}}{\sum_{i=1}^n (E_{PV,\eta=0.95})_i} \quad (31)$$

where the expected power generation of the PV solar field is represented by $E_{PV,\eta=0.95}$ and $E_{PV,\eta=f(NP)}$ utilizing constant inverter efficiency ($\eta = 95\%$) and variable inverter efficiency as specified in Eq. (31). And n stated for the number of data which equals in this case 8760. It is discovered that the average estimated error caused by the inverter efficiency is $<0.015\%$.

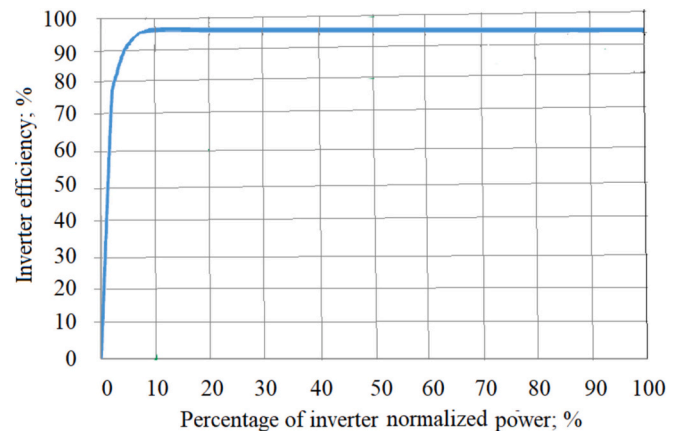


Fig. 24. Inverter efficiency variation with percentage of inverter normalized power [74].

5. Conclusions

The abundance of wind and solar energy resources in Libya, along with the availability of promising highland areas that could be used for the establishment of pumped hydropower storage (PHS) stations, are highlighted in this study and many others. As a result, this study supports the decentralization and decolonization of power generation, which is part of the Libyan government's plan to switch to renewable energy.

A 762,161 MWh storage capacity PHS, a 385 MW wind turbine farm, and a 6000 MW solar PV field make up the robust, grid-connected HRES that has been designed. The nation's yearly electricity deficit of 9,744,033 MWh can be fully met by this system. Variations in the turbine and pump's efficiencies may introduce errors of about 8 % in the turbine's energy output and 5 % in the pump's energy consumption, according to sensitivity analysis based on fluid flow rates. On the other hand, variations in inverter efficiency have a negligible impact—<0.2 %—and can therefore be disregarded in the computations.

The project's net present value (NPV) is predicted to be \$19.923 billion, based on an LCOE of \$132.1/MWh. The investment is estimated to break even in around 7.54 years. Additionally, the system is expected to avoid the emission of 11,624 kton of CO₂ each year. The yearly economic advantages include \$2.189 billion from reduced oil use, \$974.403 million from power sales, and \$665.838 million from the societal cost of CO₂. This study does not take into account the impact of weather factors (such as evaporation and precipitation) on upper reservoir water levels, nor does it consider variations in power supply or turbine performance caused by fluctuations in the upper reservoir's water flow rate, which may affect pump efficiency. These two issues are intended to be solved in future study.

6. Future work

Authors recommend future research that should focus on real-time

capabilities. As a result, they intend to demonstrate a control system that can function, observe, and react to current circumstances instantly or with little delay. Optimizing the performance and stability of renewable energy systems—which frequently struggle with variable inputs like sunlight and wind—requires this functionality. To preserve grid stability during times when renewable energy penetration is high, the hybrid system uses hydropower storage (HPS) as a buffer. When wind and solar energy surpass grid demand, pumping operations store potential energy for later use. When renewable energy production is low, the grid is stabilized by using the stored hydropower. To optimize the distribution of energy between pumping operations and direct grid feed, the authors will also present in future work predictive algorithms and real-time energy management systems. Grid reliability is further improved by ancillary services like voltage support and frequency regulation. Under conditions of fluctuating renewable energy, this integrated approach guarantees stable grid operation and effective resource utilization.

CRedit authorship contribution statement

Yasser Fathi Nassar: Writing – original draft, Methodology, Investigation, Conceptualization. **Hala Jarallah El-Khozondar:** Validation, Software, Formal analysis. **Masoud Ali Fakher:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

A.1. Fluid mechanics analysis

The diameter of the penstock (D):

$$D = 0.72 \dot{V}^{0.5} \quad (A1)$$

The head loss (h_f) is given by the Darcy-Weisbach formula:

$$h_f = \frac{8 L f \dot{V}^2}{\pi^2 g D^5} \quad (A2)$$

where f is the friction factor, L is the penstock length (m). The value of 0.2 could be set for f in the Darcy-Weisbach formula to get an initial estimation of head loss.

The friction factor (f) is determined by:

$$f = \left\{ 1.8 \log \left[\frac{6.9}{Re} + \left(\frac{\epsilon}{3.7 D} \right)^{1.11} \right] \right\}^{-2} \quad (A3)$$

where Re is the Reynold's number, ϵ is the roughness of the penstock.

The Reynold's number (Re) is determined in terms of volumetric flow rate as:

$$Re = \frac{4 \dot{V}}{\pi D \nu} \quad (A4)$$

where ν is the kinematics viscosity (m²/s).

Data availability

The datasets used during the current study available from the corresponding author on reasonable request.

References

- [1] H. El-Khozondar, J. Rifa, F. El-Batta El-Khozondar, R. Al Afif, C. Pfeifer, Microgrid for remote area in Gaza Strip powered by solar energy and olive oil mill waste, in: 12th International Conference on Sustainable Energy & Environmental Protection (SEEP 2021), Vienna, Austria, 14th - 17th September, 2021.
- [2] H. El-Khozondar, F. El-batta, Solar energy implementation at household in Gaza Strip, *Energy Sustain. Soc.* 12 (2022) 12–17.
- [3] H. El-Khozondar, F. El-batta, Hybrid energy system for Dier El Balah quarantine center in Gaza Strip, Palestine, in: 2021 International Conference on Electric Power Engineering – Palestine (ICEPE-P), 2021.
- [4] H. El-Khozondar, T. Khatib, B. Attae, R. El-Khozondar, Assessment of social acceptance of solar energy cooker in Gaza, *Sci. Rep.* 12 (2022) 17226.
- [5] IEA, "Renewable electricity 2023," [Online]. Available: <https://www.iea.org/reports/renewables-2023/electricity>.
- [6] A. Blakers, B. Lu, K. Anderson, A. Nadolny, Pumped hydro energy storage to support 100% renewable electricity, in: 2018 IEEE 7th World Conference on Photovoltaic Energy Conversion (WCPEC) (A Joint Conference of 45th IEEE PVSEC, 28th PVSEC & 34th EU PVSEC), Waikoloa, HI, USA, 2018.
- [7] Y. Nassar, H. El-Khozondar, A. Alatrash, B. Ahmed, R. Elzer, A. Ahmed, I. Imbayah, A. Alsharif, M. Khaleel, Assessing the viability of solar and wind energy technologies in semi-arid and arid regions: A case study of Libya's climatic conditions, *Applied Solar Energy* 60 (1) (2024) 149–170.
- [8] S. Mohammed, Y. Nassar, W. El-Osta, H. El-Khozondar, A. Miskeen, A. Basha, Carbon and energy life cycle analysis of wind energy industry in Libya, *Solar Energy and Sustainable Development Journal* 12 (1) (2023) 50–69.
- [9] M. Elmnifi, M. Khaleel, S. Vambol, S. Yeremenko, Y. Nassar, O. Dzhalui, Ensuring sustainability in Libya with renewable energy and pumped hydro storage, *Ecol. Quest.* 35 (3) (2024) 1–17.
- [10] N. Yasser, et al., Technical-economical-environmental analysis of strategies towards a green economy in the electricity industry sector, *Solar Energy and Sustainable Development Journal* 14 (1) (2025) 1–16.
- [11] Y. Nassar, H. El-Khozondar, N. Abohamoud, A. Abubaker, A. Ahmed, A. Alsharif, M. Khaleel, Regression model for optimum solar collectors' tilt angles in Libya, in: The 8th International Engineering Conference on Renewable Energy & Sustainability (ieCRES 2023), Gaza Strip, Palestine, May 8–9, 2023.
- [12] H. El-Khozondar, F. El-batta, Solar energy as an alternative to conventional energy in Gaza Strip: questionnaire based study, *An-Najah University Journal for Research - A (Natural Sciences)* 32 (1) (2018) 47–74.
- [13] E. Hakizimana, H. Umuhoza, E. Manishimwe, V. Kayibanda, Economic optimization of grid-connected photovoltaic solar systems in industrial energy: case study SULFO Ltd - Rwanda, *Solar Energy and Sustainable Development Journal* 13 (2) (2024) 204–229.
- [14] M. Javidan, M. Asgari, M. Gholinia, M. Nozari, A. Asgari, D. Ganji, Thermal energy storage inside the chamber with a brick wall using the phase change process of paraffin materials: a numerical simulation, *Theor. Appl. Mech. Lett.* 12 (3) (2022) 100329.
- [15] S. Shahlaei, M. Hassankolaei, MHD boundary layer of GO–H₂O nanoliquid flow upon stretching plate with considering nonlinear thermal ray and Joule heating effect, *Heat Transfer—Asian Research* 48 (8) (2019) 4152–4173.
- [16] M. Zadeh, M. Pourfallah, S. Sabet, M. Gholinia, S. Mouloudi, A. Ahangar, Performance assessment and optimization of a helical Savonius wind turbine by modifying the Bach's section, *SN Appl. Sci.* 3, no (2021) 739.
- [17] A. Mahmoud, O. Albadry, M. Mohamed, H. El-Khozondar, Y. Nassar, A. Hafez, Charging systems/techniques of electric vehicle: a comprehensive review, *Solar Energy and Sustainable Development Journal* 13 (2) (2024) 18–44.
- [18] M. Armin, M. Gholinia, Comparative evaluation of energy, performance, and emission characteristics in dual-fuel (CH₄/Diesel) heavy-duty engine with RCCI combustion mode, *Results in Engineering* 16 (2022) 100766.
- [19] A. Abodwair, M. Guneser, M. Khaleel, F. Yasser, H. El-Khozondar, A. Ibaz., Feasibility assessment of hybrid renewable energy based EV charging station in Libya, *Solar Energy and Sustainable Development Journal* 13 (2) (2024) 311–349.
- [20] F. Yasser, H. El-Khozondar, M. Khaleel, A. Ahmed, A. Alsharif, M. Elmnifi, M. Salem, I. Mangir, Design of reliable standalone utility-scale pumped hydroelectric storage powered by PV/Wind hybrid renewable system, *Energy Convers. Manag.* 322 (2024) 119173.
- [21] R. Ikreedeeh, M. Hossen, M. Tahir, 2D/2D GO-modified g-C₃N₄ nanocomposite for efficient photocatalytic CO₂ reduction to CH₄ under visible light, *Solar Energy and Sustainable Development Journal* 13 (2) (2024) 254–263.
- [22] R. Madihi, M. Pourfallah, M. Gholinia, M. Armin, A. Ghadi, Thermofluids analysis of combustion, emissions, and energy in a biodiesel (C11H22O2)/natural gas heavy-duty engine with RCCI mode (part II: fuel injection time/fuel injection rate), *International Journal of Thermofluids* 16 (2022) 100200.
- [23] S. Saeed, T. Siraj, Global renewable energy infrastructure: pathways to carbon neutrality and sustainability, *Solar Energy and Sustainable Development Journal* 13 (2) (2024) 183–203.
- [24] H. Nabi, M. Gholinia, D. Ganji, Employing the (SWCNTs-MWCNTs)/H₂O nanofluid and topology structures on the microchannel heat sink for energy storage: a thermal case study, *Case Studies in Thermal Engineering* 42 (2023) 102697.
- [25] D. Bogdanov, O. Koskinen, A. A. and C. Breyer, "Integrated renewable energy based power system for Europe, Eurasia and MENA regions," in 2016 International Energy and Sustainability Conference (IESC), Cologne, Germany, 2016.
- [26] R. Vitorino, D.A.K. Ramos, S. Paltsev, J. Reilly, Storage plants influencing the optimal hourly dispatch in hydro-thermal systems with massive integration of variable renewable energy, *Electr. Power Syst. Res.* 220 (7) (2023) 109282.
- [27] N. Naval, J. Yusta, R. Sánchez, F. Sebastián, Optimal scheduling and management of pumped hydro storage integrated with grid-connected renewable power plants, *Journal of Energy Storage* 73 (12) (2023) 108993.
- [28] F. Nassar, M. Abdunnabi, M. Sbeta, A. Hafez, K. Ali, A. Hassan, B. Belgasim, Dynamic analysis and sizing optimization of a pumped hydroelectric storage-integrated hybrid PV/Wind system: a case study, *Energy Convers. Manag.* 229 (2021) 113744.
- [29] B. Das, M. Hasan, F. Rashid, Optimal sizing of a grid-independent PV/diesel/pump-hydro hybrid system: a case study in Bangladesh, *Sustain Energy Technol Assess* 44 (4) (2024) 100997.
- [30] M. Simão, H. Ramos, Hybrid pumped hydro storage energy solutions towards wind and PV integration: improvement on flexibility, reliability and energy costs, *Water* 12 (2021) 2457.
- [31] Y. Nassar, H. El-Khozondar, G. Ghaboun, M. Khaleel, Z. Yusupov, A. Ahmed, A. Alsharif, Solar and wind atlas for Libya, *International Journal of Electrical Engineering and Sustainability (IJEES)* 1 (3) (2023) 27–43.
- [32] M. Abdunnabi, I. Rohuma, E. Endya and E. & Belal, "Review on solar water heating in Libya," *Solar Energy and Sustainable Development Journal*, vol. 7, no. 51, pp. 1–27, 2018.
- [33] K. Bakouri, T. Fohqa, O. Ahwidi, A. Abubaker, Y. Nassar, H. El-Khozondar, Learning lessons from Murzuq-Libya meteorological station: evaluation criteria and improvement recommendations, *Solar Energy and Sustainable Development Journal* 12 (1) (2023) 30–48.
- [34] S. Al-Behadili, W. El-Osta, Life cycle assessment of Dernah (Libya) wind farm, *Renew. Energy* 83 (2015) 1227–1233.
- [35] L. Rtemi, W. El-Osta, A. Attaipe, Hybrid system modeling for renewable energy sources, *Solar Energy and Sustainable Development Journal* 12 (1) (2023) 13–28.
- [36] A. Makhzom, A. Eshdok, F. Yasser, S. Alsadi, T. Fohqa, M. Salem, I. AlShareef, H. El-Khozondar, Estimation of CO₂ emission factor for Power Industry Sector in Libya, in: The 8th International Engineering Conference on Renewable Energy & Sustainability (ieCRES 2023), Gaza Strip - Palestine, May 8–9, 2023.
- [37] N. Yasser, M. Salem, K. Iessa, I. AlShareef, K. Ali, M. Fakher, Estimation of CO₂ emission factor for the energy industry sector in Libya: a case study, *Environ. Dev. Sustain.* 23 (2021) 13998–14026.
- [38] Y. Fathi, K. Aissa, S. Alsadi, Air pollution sources in Libya, *Research & Reviews: Journal of Ecology and Environmental Sciences* 6 (1) (2018) 63–79.
- [39] M. Hossain, N. Abboodi, A. Al-Fatlawi, M. El Haj, High penetration of solar photovoltaic structure on the grid system disruption: an overview of technology advancement, *Sustainability* 15 (2) (2023) 1174.
- [40] Y. Fathi, A. Salem, The reliability of the photovoltaic utilization in southern cities of Libya, *Desalination* 209 (1–3) (2007) 86–90.
- [41] Y. Nassar, S. Alsadi, Wind energy potential in Gaza Strip-Palestine state, *Solar Energy Sustain. Dev.* 7 (2) (2018) 41–57.
- [42] A. Abdalla, W. El-Osta, F. Yasser, W. Husien, H. Dekam, G. Miskeen, Estimation of dynamic wind shear coefficient to characterize best fit of wind speed profiles under different conditions of atmospheric stability and terrains for the assessment of height-dependent wind energy in Libya, *Applied Solar Energy* 59 (2023) 343–359.
- [43] L. Dinglin, Y. Chen, K. Zhang, M. Zeng, Economic evaluation of wind-powered pumped storage system, *Systems Engineering Procedia* 4 (2012) 107–115.
- [44] Y. Nassar, S. Alsadi, G. Miskeen, H. El-Khozondar, N. Abohamoud, Atlas of PV solar systems across Libyan territory, in: 2022 International Conference on Engineering & MIS (ICEMIS), Istanbul, Turkey, 04–06 July, 2022.
- [45] T. Stehly, P. Duffy, D. Mulas, 2022 Cost of Wind Energy, National Renewable Energy Laboratory-NREL, December 2023.
- [46] A. Hafez, Y. Nassar, M. Hamdan, S. Alsadi, Technical and economic feasibility of utility-scale solar energy conversion systems in Saudi Arabia, Iran, *J. Sci. Technol. Trans. Electr. Eng.* 44 (1) (2019) 213–225.
- [47] M. Abdunnabi, N. Etiab, Y. Nassar, H. El-Khozondar, R. Khargotra, Energy savings strategy for the residential sector in Libya and its impacts on the global environment and the nation economy, *Adv. Build. Energy Res.* 17 (4) (2023) 379–411.
- [48] M. Eteriki, W. El-Osta, Y. Nassar, H. El-Khozondar, Effect of implementation of energy efficiency in residential sector in Libya, in: The 8th International Engineering Conference on Renewable Energy & Sustainability, Gaza Strip, Palestine, May 8–9, 2023.
- [49] L. Al-Ghussain, A. Ahmad, A. Abubaker, M. Mohamed, An integrated photovoltaic/wind/biomass and hybrid energy storage systems towards 100% renewable energy microgrids in university campuses, *Sustain Energy Technol Assess* 46 (2021) 101273.
- [50] S. Ghandehariun, A. Ghandehariun, N. Ziabari, Complementary assessment and design optimization of a hybrid renewable energy system integrated with open-loop pumped hydro energy storage, *Renew. Energy* 227 (6) (2024) 120557.
- [51] Y. Nassar, H. El-Khozondar, W. El-Osta, S. Mohammed, M. Elnaggar, M. Khaleel, A. Ahmed, A. Alsharif, Carbon footprint and energy life cycle assessment of wind energy industry in Libya, *Energy Convers. Manag.* 300 (1) (2023) 117846.
- [52] H. El-Khozondar, F. El-batta, R. El-Khozondar, Y. Nassar, M. Alramlawi, S. Alsadi, Standalone hybrid PV/Wind/Diesel electric generator system for a COVID-19 Quarantine Center, *Environ. Prog. Sustain. Energy* (2022) 1–18.
- [53] Y. Nassar, *Solar Energy Engineering, Active Applications*, Sebha, Sebha University, Libya, 2006.

- [54] Y. Fathi, S. Alsadi, Assessment of solar energy potential in Gaza Strip-Palestine, *Sustain Energy Technol Assess* 31 (2019) 318–328.
- [55] IRENA, Renewable Power Generation Costs in 2022, International Renewable Energy Agency, Abu Dhabi, 2023.
- [56] H. El-Khozondar, et al., Photovoltaic solar energy for street lighting: a case study at Kuwaiti Roundabout, Gaza Strip, Palestine, *Power Engineering and Engineering Thermophysics* 3 (2) (2024) 77–91.
- [57] M. Elnaggar, H. El-Khozondar, W. Salah, N. Yasser, M. Bashir, Assessing the techno-enviro-economic viability of wind farms to address electricity shortages and Foster sustainability in Palestine, *Results in Engineering* 24 (2024) 103111.
- [58] H. Hammada, A sudden increase in fuel costs for power plants in Libya, *The Libyan Position* (191) (2024).
- [59] A. Risso, F. Canales, A. Beluco, and E. Rossini, "A PV wind hydro hybrid system with pumped storage capacity installed in Linha Sete, Aparados da Serra, southern Brazil," Ch. 10 in *Modeling and Dynamic Behaviour of Hydropower Plants Book*, 2024.
- [60] N. Yimen, O. Hamandjoda, I. Meva'a, B. Ndzana, J. Nganhon, "Analyzing of a photovoltaic/wind/biogas/pumped-hydro off-grid hybrid system for rural electrification in Sub-Saharan Africa—case study of Djoundé in Northern Cameroon." *Energies* vol. 11, p. 2644, 2018.
- [61] M. Cruz, I. Yahyaoui, R. Fiorotti, M. Segatto, A. Atieh, H. Rocha, Sizing and energy optimization of wind/floating photovoltaic/hydro-storage system for Net Zero Carbon emissions in Brava Island, *Renewable Energy Focus* 47 (12) (2023) 100486.
- [62] S. Guo, A. Kurban, Y. He, F. Wu, H. Pei, G. Song, Multi-objective sizing of solar-wind-hydro hybrid power system with doubled energy storages under optimal coordinated operational strategy, *Journal of Power and Energy Systems* 9 (6) (2023) 2144–2155.
- [63] X. Xu, W. Hu, D. Cao, Q. Huang, C. Chen, Z. Chen, Optimized sizing of a standalone PV-wind-hydropower station with pumped-storage installation hybrid energy system, *Renew. Energy* 147 (1) (2020) 1418–1431.
- [64] Y. Xu, Y. Lang, B. Wen, X. Yang, An innovative planning method for the optimal capacity allocation of a hybrid wind–PV–pumped storage power system, *Energies* 12 (2019) 2809.
- [65] A. Menesy, S. Hamdy, I. Habiballah, H. Masrur, K. Khan, M. Khalid, Optimal configuration of a hybrid photovoltaic/wind turbine/biomass/hydro-pumped storage-based energy system using a heap-based optimization algorithm, *Energies* 16 (9) (2023) 3648.
- [66] T. Ma, H. Yang, L. Lu, J. Peng, Pumped storage-based standalone photovoltaic power generation system: modeling and techno-economic optimization, *Appl. Energy* 137 (2015) 649–659 (ISSN 0306-2619).
- [67] B. Kommula, H. Song, L. Chen, C. Xu, Efficient energy management of hybrid renewable energy sources-based smart-grid system using a hybrid IDEA-CFA technique, *International Transactions on Electrical Energy Systems* 31 (5) (2021) e12833.
- [68] H. Koli, M. Chawla, Design analysis of pv-wind energy system with pumped hydro storage using HOMER Pro, *International Journal of Recent Technology and Engineering (IJRTE)* 9 (3) (2020) 502–505.
- [69] A. Al-akayshee, O. Kuznetsov and H. Sultan, "Optimization and operation of stand-alone hybrid PV/biomass/hydroelectric pumped storage energy system in Iraq," 2021 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering (ElConRus), St. Petersburg, Moscow, Russia, 2021, pp. 1365–1370.
- [70] H. Al-Masri, O. Dawaghreh, S. Magableh, Optimal configuration of a large scale on-grid renewable energy systems with different design strategies, *J. Clean. Prod.* 414 (8) (2023) 137572.
- [71] J. Jurasz, J. Mikulik, M. Krzywda, B. Ciapała, M. Janowski, Integrating a wind- and solar-powered hybrid to the power system by coupling it with a hydroelectric power station with pumping installation, *Energy* 144 (2018) 549–563.
- [72] M. Simão, R. Helena, Hybrid pumped hydro storage energy solutions towards wind and PV integration: improvement on flexibility, reliability and energy costs, *Water* 12 (9) (2020) 2457.
- [73] R. Dufo-López, J. Lujano-Rojas, Optimisation of utility-scale PV–wind systems with pumped hydro storage, *Appl. Sci.* 14 (2024) 7033.
- [74] B. Yang, D. Li, X. Fu, H. Wang, R. Gong, Parameter impact and sensitivity analysis of a pumped hydro compressed air energy storage system, *Appl. Therm. Eng.* 246 (2024) 122951.
- [75] R. Dufo-López and M. Lujano-Rojas, "Simulation and optimisation of utility-scale PV–wind systems with pumped hydro storage." *Applied Sciences*. vol. 14, no. 16, p.7033, 224.
- [76] F. Yasser, S. Alsadi, H. El-Khozondar, M. Ismail, M. Al-Maghalseh, T. Khatib, J. Sa'ed, M. Mushtaha, T. Djeraji, Design of an isolated renewable hybrid energy system: a case study, *Materials for Renewable and Sustainable Energy* 11 (2022) 225–240.